

Admissibility, damping margin and island grid strength in the IEEE-14 grid-forming switching study

Why a switching event promotes one or two converters, and three proposed switching indices

In-house power system analysis toolkit

Abstract

The IEEE-14 study network carries one synchronous machine at bus 1 and four dual-mode converters at buses 2, 3, 6 and 8, each able to run grid-following (GFL) or grid-forming (GFM). When the machine trips, a supervisor decides how many converters to promote and which ones. This report establishes, from the delivered code and the delivered result caches, that the promotion count is not chosen by a stability objective and not by “more is safer”: it is the *minimum-cardinality strict superset* of the currently forming set that appears in a pre-computed, authenticated admissibility table, and that table is *not monotone* in the count. Of the fifteen possible forming sets, only seven are admissible; *every* three-converter set has no equilibrium at all (coupled-Newton residual $3.28\text{--}3.61 \times 10^2$). The acceptance criterion is a robust damping-ratio floor $\zeta_{\min} = 0.05$ evaluated at three frozen finite-difference steps, and the all-four set is admissible but is the least damped of the seven ($\zeta_{\text{worst}} = 0.1223$ against $0.2549\text{--}0.2676$). The island short-circuit ratio at the converter buses stays between 2.30 and 2.73 in every islanded window — below the WECC strong-grid floor of 3 on 98.61% of all converter-samples, and below it on 100% of the post-trip and post-reclose windows. The delivered adaptive chronology promotes one unit at a time, holds two for most of the islanded window and reaches four only when nothing smaller is admissible. Finally, the report measures five failure modes of the present two-term voltage/frequency switching index on the delivered run — among them that its frequency half is *identical* across all four converters for all 3679 samples — and designs three additional indices that each enter their own non-tradeable condition rather than a re-weighted scalar.

Contents

1	Scope, questions and evidence base	3
1.1	The five questions	3
1.2	What “proof” means here, and what it does not	3
1.3	Artefacts	5
1.4	The delivered result, for orientation	6
2	The studied system and what the supervisor is for	6
2.1	The network and the resources	6
2.2	What the machine carries, and what the trip hands over	6
2.3	The delivered switching history	7

3	The decision pipeline	7
3.1	Stage 1 — when support is requested	8
3.2	Stage 2 — which converters	9
3.3	Stage 3 — how the transition is applied	9
4	The admissibility enumeration	9
4.1	What the enumeration proves	11
5	The damping margin: how it is computed, and why this value	11
5.1	The computation, step by step	11
5.2	Why a damping <i>ratio</i> rather than a decay rate	12
5.3	Why 0.05	12
5.4	The measured margins	13
6	Why a switching event promotes one or two, not four	13
6.1	The mechanism	13
6.2	Correcting the premise, and then answering it	14
6.3	Why the release is faster than the promotion	15
7	Island short-circuit strength after the machine trips	15
7.1	How the project defines SCR	15
7.2	The gate is not applicable to this model family	15
7.3	The measured island SCR	16
7.4	Reading the number	16
8	What happens if all four converters form	17
8.1	The all-four arm does not survive the island	17
8.2	Why an admissible set can still fail	18
8.3	Why all four is worse — three claims, each with its scope	18
8.4	Is a strong grid not a good thing?	19
9	Where the present two-term index fails	20
9.1	The frequency term carries no device identity	20
9.2	It saturates before it is informative	20
9.3	It cannot see a device state, and the all-four failure lives there	21
9.4	The PLL lock margin is measured, published, and not used	21
9.5	Current headroom is measured, published, and not used	21
9.6	Summary of the gap	21
10	Two additional switching indices: RoCoF and RoCoV	21
10.1	The design constraint that shapes both	21
10.2	Index A — RoCoF: the per-device rate of change of frequency	22
10.3	Index B — RoCoV: the rate of change of voltage	23
10.4	What is deliberately not proposed	24
11	What this report does not claim	25
12	Reproduction	25

1 Scope, questions and evidence base

1.1 The five questions

1. **Promotion count.** In the latest model code, why does a switching event not place every converter in grid-forming mode, but select only one or two?
2. **Damping margin.** From what quantity is the margin computed, how is it computed, and why is that the quantity that is used?
3. **Island grid strength.** What is the short-circuit ratio (SCR) at the converter buses once the synchronous machine is tripped out?
4. **All four forming.** What happens if all four converters form, why is that undesirable, and is a strong grid not a good thing?
5. **Switching index.** The index that drives switching today uses only voltage and frequency. That is judged too narrow and capable of producing false decisions. Three further indices are designed.

1.2 What “proof” means here, and what it does not

This report proves statements of two different kinds, and keeps them apart.

Model-internal statements are proved by reading the production code that the delivered run executes. “The promotion is the minimum-cardinality strict superset” is a statement of this kind: it is a property of `select_support_augmentation_candidate.m` and of the table the run consumes, and it holds for every run of this code, not only for the delivered one.

Trajectory statements are proved by measurement on a named artefact. “The island SCR is 2.3 once the machine is out” is a statement of this kind: it is true of `adaptive_250s.mat` and of the network and dispatch that cache was built from, and it is not claimed to be universal.

No statement in this report is proved by argument alone. Where a mechanism is offered as an explanation rather than as a measurement, it is marked as an explanation at the point of use, and Section 12 lists what is *not* claimed.

1.3 Artefacts

Artefact	Role in this report
+stability/ts_simulate_ibr_hybrid.m	the supervisor: severity formation, hysteresis, dwell, and the call into candidate selection and the support transaction.
+stability/select_support_augmentation_candidate.m	the ranking rule that turns “severity says more support is needed” into one named forming set.
+stability/ibr_candidate_evaluate.m	the acceptance criterion: equilibrium, KCL, full-KCL small-signal spectrum and the robust damping-ratio gate.
+stability/ibr_scr_metrics.m	the short-circuit-ratio metric and its gate.
+stability/agsi_reference_terms.m	the reference sub-indices published beside the run, including the three the decision does not consume.
+cases/case_ieee14_1sg_4ibr_auto_vsg.m	the case, its dispatch contract and its frozen selector contract (case_data.selector, :168--174).
run_ieee14_gfm_lock_comparison.m	the six-arm study and the shared option set every arm starts from.
ieee14_gfm_lock_compare_zeta/adaptive_250s.mat	the delivered 250 s adaptive chronology. Every trajectory number below is read from this file.

1.4 The delivered result, for orientation

Quantity	Value	Source
horizon reached	250.000000 s	result.t(end)
accepted samples	3679	numel(result.t)
rejected steps / attempts	161 / 3824	rejected_steps, ts_step_attempts
max accepted KCL residual	9.972081×10^{-9}	max(accepted_residual_per_step)
terminal f_{COI}	60.000001128 Hz	coi_frequency_Hz(end)
SG reclose	159.2519 s, SUCCESS	actual_reclose_time
reference handback complete	173.1271 s, C1_COMPLETE	handback_complete_time

Table 2: The delivered adaptive chronology. These seven values are the project’s regression gate for this result set.

2 The studied system and what the supervisor is for

2.1 The network and the resources

The case is `case_profile = 'eecon49_figure4'` on the IEEE-14 network, resolved to `ibr_model_id = 'eecon49_dual'`. The resource table is fixed and its order is the only identity the selector uses:

Index	Id	Bus	Type	Base (MVA)	Initial	Model
1	SG1	1	synchronous	615	online	sg_emf6
2	IBR2	2	converter	100	GFL	eecon49_dual
3	IBR3	3	converter	100	GFL	eecon49_dual
4	IBR6	6	converter	100	GFL	eecon49_dual
5	IBR8	8	converter	100	GFL	eecon49_dual

Table 3: The resource table. Bus numbers and device names are never decision keys; resource-table indices are.

The machine contributes $H = 2.5$ s and $D = 1.0$ pu in the delivered option set. Each converter carries a filter $L_f = 0.15$ pu, $R_f = 0.015$ pu, a DC link $C_{dc} = 0.10$, and a current ceiling $I_{\text{max}} = 1.2$ pu on the common 100 MVA base. The grid-forming branch is a virtual synchronous machine with swing coefficient $M = 0.08$ and damping coefficient $D_v = 20.0$; the grid-following branch is a PLL-based current controller with $k_{p,\text{PLL}} = 1.20$, $k_{i,\text{PLL}} = 5.00$. The same D_v coefficient fixes both the P – f droop and the swing damping, which is a property of this single-coefficient VSG and not a modelling oversight.

2.2 What the machine carries, and what the trip hands over

At the pre-event power flow the machine supplies 1.3319 pu and the four converters together supply 1.3176 pu, against a total load of 2.59 pu of active power and 0.735 pu of reactive power. The case’s own dispatch contract records the consequence of opening the machine breaker as `deficit_MW = 133.1904` and `lost_sg_Q_MVAr = -19.5666`. **[measured]**

Opened at $t = 20$ s, that breaker leaves an energized island whose voltage is formed by converters alone, and it obliges the supervisor to answer two separate questions: *when* is more forming support required, and *which* converters should provide it. The code answers them in two different places, and that separation is the subject of Section 3.

2.3 The delivered switching history

t (s)	event	forming set	transition	variant
0.0000	initial	{}	all GFL, SG online	—
20.0000	SG trip	{2}	IBR2 $\rightarrow$ GFM	—
22.0887	support augment	{2, 4}	+IBR6	unconditioned
25.3026	support release	{2}	−IBR6	unconditioned
50.0000	load step +20%	{2}	—	—
51.3403	support augment	{2, 4}	+IBR6	unconditioned
53.3795	support augment	{2, 3, 4, 5}	+IBR3, +IBR8	conditioned
85.0000	fault, bus 9	{2, 3, 4, 5}	—	—
85.1500	fault cleared	{2, 3, 4, 5}	—	—
110.0000	line 6–13 trip	{2, 3, 4, 5}	—	—
145.0000	topology restored, reclose requested	{2, 3, 4, 5}	—	—
146.2782	support release	{2}	−IBR3, IBR6, IBR8	unconditioned
148.2811	support augment	{2, 4}	+IBR6	unconditioned
152.4017	support release	{2}	−IBR6	unconditioned
159.2519	SG reclose	{2}	SUCCESS	—
173.1271	handback, IBR2 $\rightarrow$ GFL	{}	—	—

Table 4: Every mode transaction of the delivered adaptive chronology, read from `result.device_modes_history` and `result.event_log`. **[measured]**

Read the “transition” column as the answer to Q1 in its most literal form. Five augment or release events occur in 250 s, and *every one of them moves by one or two units*. The set never jumps from {2} to {2, 3, 4, 5} in a single transaction. Section 6 shows that this is structural, not incidental.

3 The decision pipeline

The supervisor is three stages in series, and each stage can only refuse. No stage can overrule a refusal by another.

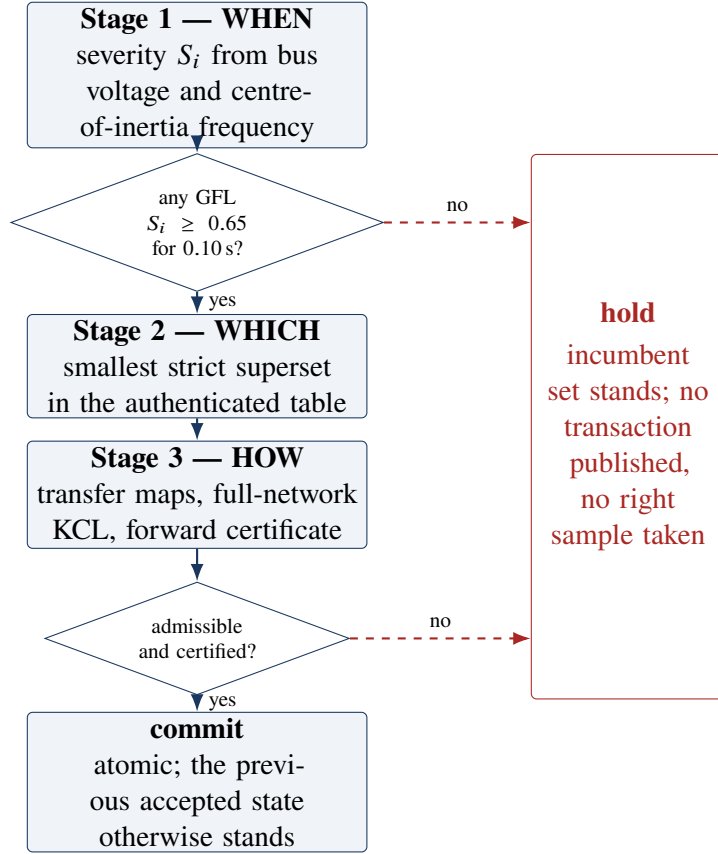


Figure 1: The three stages. Stage 1 is a new decision every sample; stages 2 and 3 are lookups and tests against evidence frozen before the run started.

3.1 Stage 1 — when support is requested

For every online switchable converter the driver forms

$$S_i = \text{sat}\left(\frac{1}{2}J_{V,i} + \frac{1}{2}J_f\right), \quad J_{V,i} = \frac{|V_i - V_{i,\text{healthy}}|}{0.10}, \quad J_f = \frac{|f_{\text{COI}} - 60|}{0.50}, \quad (1)$$

with $\text{sat}(x) = \min(1, \max(0, x))$ (ts_simulate_ibr_hybrid.m:3631), bases 0.10 pu and 0.50 Hz fixed at :1675--1677, and $V_{i,\text{healthy}}$ the pre-event power-flow voltage of that converter's own bus. The promotion and release conditions are

$$\text{augment} = |\{i \in \text{GFL} : S_i \geq 0.65\}| > 0, \quad \text{release} = \forall i, S_i < 0.35, \quad (2)$$

sustained for $T_{d,\text{on}} = 0.10$ s and $T_{d,\text{off}} = 1.00$ s respectively (:852--900), with the thresholds set at :1664--1667. Three properties of (2) matter later:

- the augment term is an *any* over the *followers*. The index therefore asks “is a grid-following converter under stress?”, not “is the system stable?” and not “which converter is the cause?”.
- the release term is an *all*. Hysteresis is asymmetric on purpose: support is added on the first follower in trouble and withdrawn only when every converter is comfortable for a full second.
- no term of (1) is a stability quantity. Feasibility, small-signal margin, current limits and reference ownership are all elsewhere, and Section 9 argues that this is correct — but that the *set* of quantities entering Stage 1 is too small.

3.2 Stage 2 — which converters

Stage 1 produces a direction, never a set. The set comes from a table of candidate configurations computed before the simulation starts (`ibr_selector_table`), and the runtime rule is `select_support_augmentation_candidate`:

Proposition 1 (Minimum-cardinality admissible superset). *Let $\mathcal{F}$ be the forming set at the decision instant and $\mathcal{T}$ the set of table rows with *feasible* and *ready_to_commit* both true. The returned destination is the row $c \in \mathcal{T}$ minimising, in order: (i) $|\text{selected}(c)|$; (ii) $-\text{margin}(c)$; (iii) the selected index tuple lexicographically; (iv) the reference index — subject to $\mathcal{F} \subsetneq \text{selected}(c)$, i.e. the destination must be a strict superset of the current set (`select_support_augmentation_candidate.m:54--58, 101--112`). If no such row exists the function returns `found=false` and the supervisor holds its incumbent set; it never fabricates a candidate and never relaxes a gate.*

Two consequences are immediate and are the direct answer to Q1.

1. **The cardinality objective is a minimum, not a maximum.** Nothing in Proposition 1 rewards adding converters. A three-unit destination is chosen over a two-unit destination only when no two-unit strict superset is admissible.
2. **The choice is not a free re-selection.** Because the destination must contain the current set, the supervisor cannot answer “the island needs four formers” by promoting four units at once from one former; it can only add to what it has, and it adds as few as the table permits.

3.3 Stage 3 — how the transition is applied

The mode change passes through device-owned transfer maps, then a full-network Kirchhoff solve at the new operating point, then — because the option is enabled in this study — a forward simulation of the would-be committed state with the same integration kernel over a horizon derived from the destination’s own settling time. A refusal is atomic: no right sample is published, the previous accepted state stands, and a named identifier plus a lockout is recorded. Over the delivered 250 s the certificate was consulted at four transactions, and it changed the outcome at one of them (Table 4, $t = 53.3795$: the untouched arrival was rejected and the conditioned variant committed at a peak excursion of 15.495°). **[measured]**

4 The admissibility enumeration

Stage 2 has nothing to choose from unless a table exists, and the table is not a formality. Five converters, four of them switchable, admit $2^4 - 1 = 15$ non-empty forming sets. The project computes every one of them.

Set	Ref	ω (rad/s)	margin	ζ_{worst}	Outcome
{5}	5	-0.155437	+0.055437	0.2676	admissible
{4}	4	-0.252025	+0.152025	0.2629	admissible
{3}	3	-0.316711	+0.216711	0.2618	admissible
{2}	2	-0.328014	+0.228014	0.2569	admissible
{2, 4}	2	-0.561723	+0.461723	0.2549	admissible
{3, 4}	3	-0.568138	+0.468138	0.2560	admissible (most damped)
{2, 3}	—	—	—	—	no equilibrium: IBR6 violates a limit
{2, 5}	—	—	—	—	no equilibrium: IBR6 violates a limit
{3, 5}	—	—	—	—	no equilibrium: IBR6 violates a limit
{4, 5}	—	—	—	—	no equilibrium: IBR3 violates a limit
{2, 3, 4}	—	—	—	—	no equilibrium, residual 3.279×10^2
{2, 3, 5}	—	—	—	—	no equilibrium, residual 3.605×10^2
{2, 4, 5}	—	—	—	—	no equilibrium, residual 3.561×10^2
{3, 4, 5}	—	—	—	—	no equilibrium, residual 3.346×10^2
{2, 3, 4, 5}	2	-0.482909	+0.382909	0.1223	admissible, least damped

Table 5: The complete SG-off admissibility enumeration, read from `summary.shared.sg_off_candidates` in `ieee14_gfm_lock_compare_zeta/summary.mat`. Seven of fifteen sets are admissible. This is a *base-load table, frozen before candidate evaluation*; the load-level sensitivity of Section 8 is a separate measurement at +20% load and does not modify these rows. ω is the least damped member of the frozen finite-difference set, $\omega = \max(\text{cand.fd_omegas})$ (`ibr_candidate_evaluate.m:447`), not the value at a single step; $\text{margin} = -\omega - \gamma_{\text{req}}$; ζ_{worst} is the worst damping ratio over the physical decision spectrum, minimised over the same three finite-difference members. **[measured]**

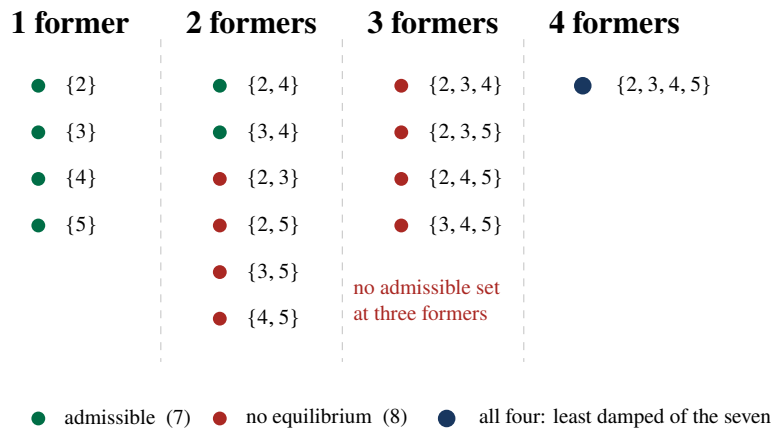


Figure 2: The admissibility map. Feasibility is *not* monotone in the number of forming converters: every admissible set sits below or above the three-unit row, never on it.

4.1 What the enumeration proves

1. **“More forming units” is not a monotone improvement, and the project measured this rather than assuming it.** The middle of the map is empty. Three forming converters is not a compromise between two and four; it is a configuration for which the coupled Newton finds no operating point at all, at any of the four ways of choosing the three.
2. **The reason is the sharing problem, not the number.** Each of the eight infeasible rows fails for a mechanical reason: either a converter left in grid-following is asked to violate an operating limit ($\{2, 3\}$, $\{2, 5\}$, $\{3, 5\}$, $\{4, 5\}$), or the four-way power balance has no solution (the four three-unit sets). In both families the failure is reported by name, not by a low margin.
3. **The all-four set is admissible but is the outlier.** Its worst damping ratio, 0.1223, is less than half of every other admissible set’s. Section 5 shows where that number comes from and why it is the right thing to gate on.

Remark 1 (Why a pre-computed table rather than an on-line search). A candidate is admitted only after its own equilibrium is solved and its own full-KCL spectrum is assembled at three perturbation sizes. That costs seconds per candidate and must not sit inside an accepted time step. Pre-computing the whole universe also makes the runtime decision a pure lookup of evidence the run can no longer influence, which is what allows Proposition 1 to be stated about the code rather than about a particular trajectory.

5 The damping margin: how it is computed, and why this value

5.1 The computation, step by step

For each candidate set the evaluator (`+stability/ibr_candidate_evaluate.m`) runs the following chain. No step is skipped, and a failure at any step stops the evaluation with a named identifier.

1. **Freeze the criterion.** $\zeta_{\min}$ is read from the case’s own contract, `case_data.selector.zeta_min_damping`, not from the caller, and is validated to lie in $(0, 1)$ (:83--93). For this case $\zeta_{\min} = 0.05$.
2. **Solve the destination equilibrium.** The candidate’s exact mode assignment is built into the device set and the hybrid state, then solved by the coupled Newton with tolerance 10^{-8} and 300 iterations (:314, :326). The machine breaker is held open throughout, because the selector owns the post-trip configuration contract and evaluating a candidate with the machine closed would certify a different physical system (:220--228).
3. **Enforce the network equations.** The physical all-row KCL residual must satisfy $\|\cdot\| \leq 10^{-6}$ (:340--345). Measured residuals for the admissible rows are 3.6×10^{-15} to 1.4×10^{-14} . **[measured]**
4. **Assemble the decision spectrum.** The full-KCL composite state-space model is built at the accepted equilibrium with the exact input u_{eq} , event context and active-state index set (:355--395). The network-angle gauge coordinate is removed by a centre-of-inertia T -matrix, and no eigenvalue is deleted after the eigenvalue call (:398--400).

5. **Damping ratio per mode.** Each eigenvalue of the physical decision spectrum is reduced to

$$\zeta_i = \frac{-\operatorname{Re} \lambda_i}{|\lambda_i|}. \quad (3)$$

A real negative root gives $\zeta = 1$ and can therefore never be the limiting mode. A root *at* the origin has an undefined ratio and is mapped to $-\infty$ rather than skipped, so a marginal mode fails any positive floor instead of quietly certifying; an empty spectrum likewise returns $-\infty$, because no evidence is not evidence of damping (:492--516).

6. **Worst over the spectrum, not rightmost.** The reported quantity is the minimum of (3) over the whole decision spectrum. This is a deliberate departure from reading the margin off $\max \operatorname{Re} \lambda$: the worst-damped mode is generally not the rightmost one, and reading the ratio off the rightmost eigenvalue would miss it (:402--410).
7. **Repeat at three perturbation sizes.** The whole assembly is redone at finite-difference steps $\varepsilon \in \{1.5, 3, 6\} \times 10^{-6}$ (the audited base times 0.5, 1, 2) and the test is applied at *every* member (:412--448).
8. **The gate.** A candidate is admissible when

$$\forall \varepsilon \in \mathcal{E} : \quad \max \operatorname{Re} \lambda(\varepsilon) < 0 \quad \text{and} \quad \min_i \zeta_i(\varepsilon) \geq \zeta_{\min}, \quad (4)$$

that is, all three perturbation sizes must agree both on stability and on the ratio floor. Disagreement between them fails closed. For the delivered table all three agree, and the reported ζ_{worst} is the worst of the three.

5.2 Why a damping *ratio* rather than a decay rate

The code retains a second, older quantity: $\gamma_{\text{req}} = 0.1$ rad/s, the reference decay rate, from which

$$\text{margin} = -\max \operatorname{Re} \lambda - \gamma_{\text{req}}. \quad (5)$$

It is used, and it is used only as the *ordering key* among admissible candidates (:452--462). The distinction is not cosmetic.

An absolute-rate floor asks $\operatorname{Re} \lambda \leq -\gamma$ and is therefore frequency-dependent in effect: the same γ is a mild requirement for a fast mode and an unreachable one for a slow mode. A ratio floor asks the mode to decay within ζ^{-1} of a radian and is invariant under a change of timescale. The two families coincide at exactly one frequency,

$$f^* = \frac{\gamma_{\text{req}}}{2\pi\zeta_{\min}} = \frac{0.1}{2\pi(0.05)} = 0.3183 \text{ Hz}, \quad (6)$$

which is the value the case contract records as `equivalence_frequency_Hz`. Away from f^* they are different requirements (:402--410). This plant carries modes from roughly 0.02 Hz — the island's electromechanical swing, slowed by the converters' small inertia — up to the current-loop band, a span of more than three decades, so no single absolute rate is a meaningful floor across it.

5.3 Why 0.05

The value is declared by the case, not chosen by the report:

```

acceptance_criterion      'damping_ratio_floor'
derivation                '5% damping at the 1Hz electromechanical mode:
                           zeta >= 0.05 for every mode of the physical decision spe
gamma_req_role            'candidate ordering key and reference decay rate; NOT th
frozen_before_candidate_eval  1

```

One apparent tension in that block should be resolved rather than left for the reader. The *derivation* names a 1 Hz electromechanical mode, while *equivalence_frequency_Hz* is 0.3183 Hz. These are not two claims about the same thing: 1 Hz is the mode at which the classical 5% criterion is quoted, and 0.3183 Hz is the frequency at which a 5% ratio and a 0.1 rad/s rate impose the same requirement, by (6). The case file therefore records the origin of the number and the crossover of the two criterion families separately, and both are needed to read the contract.

The provenance is thus threefold and should be stated plainly. The *number* 0.05 is the classical electromechanical damping criterion, used here as a case-level acceptance floor rather than as a simulation target. The *scope* — “every mode of the physical decision spectrum, at every perturbation size” — is the project’s own strengthening: a fleet of modes with one barely damped member is not rescued by the others being comfortable, and a margin that exists at one finite-difference step but not at the next is not a margin. The *frozen* flag makes the criterion a property of the case file, fixed before any candidate is evaluated, so it cannot be re-tuned to fit an outcome. This is the mechanism by which a candidate can be admitted at $\text{margin} < 0$ while being admissible: margin is a ranking score against a reference rate, and the gate is the ratio.

Remark 2 (What the gate is not). Admissibility is a property of an *equilibrium*, established by linearisation about it. It says nothing about whether the trajectory arriving at that equilibrium is inside its region of attraction. Section 8 shows a measured case in which the equilibrium was admissible and the arrival was not.

5.4 The measured margins

Table 5 gives all seven. Three observations are worth stating explicitly.

1. **Every admissible set clears the floor comfortably.** The smallest worst ratio is 0.1223, which is $2.45\times$ the floor of 0.05. The binding constraint in this case is therefore not (4) at all; it is the existence of an equilibrium at all, which removes 8 of 15 sets.
2. **The damping ratio and the decay rate disagree, and the gate uses the ratio.** On ω , the ordering is $\{3, 4\} > \{2, 4\} > \{2, 3, 4, 5\} > \{2\} > \dots$, so the all-four set is the *second* fastest-decaying admissible set. On ζ_{worst} it is the worst by a factor of two. The gate follows ζ_{worst} .
3. **The two-unit sets are the small-signal optimum, and $\{3, 4\}$ is the best of them.** $\{3, 4\}$ has both the most negative ω (-0.568138) and a top-tier ratio (0.2560). This is why the pinned-arm study uses $\{3, 4\}$ as the fixed-configuration rival to the adaptive policy.

6 Why a switching event promotes one or two, not four

6.1 The mechanism

Proposition 1 fixes the count. Work the delivered timeline through it.

t (s)	incumbent $\mathcal{F}$	candidate supersets	admissible	chosen
22.0887	{2}	{2, 3}, {2, 4}, {2, 5}, {2, 3, 4}, {2, 3, 5}, {2, 4, 5}, {2, 3, 4, 5}	{2, 4}, {2, 3, 4, 5}	{2, 4}
51.3403	{2}	(as above)	{2, 4}, {2, 3, 4, 5}	{2, 4}
53.3795	{2, 4}	{2, 3, 4}, {2, 4, 5}, {2, 3, 4, 5}	{2, 3, 4, 5} only	{2, 3, 4, 5}
148.2811	{2}	(as 22.0887)	{2, 4}, {2, 3, 4, 5}	{2, 4}

Table 6: Every augment of Table 4 resolved through Proposition 1 using the admissibility of Table 5.

At $t = 22.0887$ s and $t = 51.3403$ s two destinations exist and the smaller one wins on criterion (i). At $t = 53.3795$ s — which is after the +20% load step, with the island carrying the machine’s 133.19 MW deficit — the two-unit supersets {2, 3, 4} and {2, 4, 5} are among the eight rows with no equilibrium, so the only admissible destination is the all-four set and criterion (i) has nothing to choose. *Four forming converters is not a policy preference; it is the residue of the enumeration after the load step.*

The four-unit promotion happens because at +20% load nothing smaller can be certified — not because four was judged better than two.

6.2 Correcting the premise, and then answering it

The question as posed — “it never goes to all four, only one or two” — is *partly* true of the delivered chronology and sharply true of the scenario suite. Both halves should be on the record.

Arm	horizon	peak forming	forming at end	reclose
chronology, adaptive	250 s	4	0	SUCCESS 159.2519 s
suite, sg_load_step30	150 s	4	4	SUCCESS 81.1500 s
suite, sg_fault_bus9	150 s	2	0	SUCCESS 81.8627 s
suite, line_fault_9_14	150 s	2	0	SUCCESS 92.4126 s
suite, former_outage	150 s	2	2	SUCCESS 90.3374 s
suite, sg_fault_cycle160	160 s	2	0	SUCCESS 102.3627 s

Table 7: Peak forming count per delivered arm, read from each arm’s `device_modes_history`. Four of the five suite arms never exceed two forming converters; the two arms that reach four are the two that carry a large load step, for the reason in Table 6. **[measured]**

So the accurate statement is:

- *Per event*, the promotion is one or two units, always, structurally, for the reason proved above. In the delivered chronology the forming count takes the values $1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 1$ across the islanded window, with durations 36.98 s, 9.37 s and 92.90 s for one, two and four formers respectively. **[measured]**
- *Per arm*, four of the five delivered suite arms peak at two. The delivered 250 s chronology and sg_load_step30 reach four, and both are the arms with a heavy load step.
- *In no arm* does the supervisor jump from one former straight to four in a single transaction, because no such transaction exists in the table’s strict-superset relation over one step.

6.3 Why the release is faster than the promotion

The asymmetry in (2) ($T_{d,on} = 0.10$ s against $T_{d,off} = 1.00$ s) is deliberate: add support on the first evidence, withdraw it only after a second of calm. The measured cost is visible at $t = 146.2782$ s, where three devices return to grid-following 1.28 s after the reclose request, and the island runs on one former again from 152.4017 s to the reclose at 159.2519 s. That is the design working: the supervisor is willing to run a thin island while the machine is being pulled into phase, because a transition during the synchronising window is itself a risk. **[measured]**

7 Island short-circuit strength after the machine trips

7.1 How the project defines SCR

+stability/ibr_scr_metrics.m defines, per online converter,

$$SCR_i = \frac{S_{sc,i}}{S_{rated,i}}, \quad S_{sc,i} = \frac{|V_i|^2}{|Z_{th,ii}|} S_{base}, \quad Z_{th,ii} = [\mathbf{Y}^{-1} \mathbf{e}_i]_i, \quad (7)$$

with $\mathbf{Y}$ built from the branch data — including tap and phase shift, and bus shunts, but *not* load admittance, so that the result measures network strength rather than loading — and with reference buses treated as grounded so that the Thevenin equivalent is taken with the voltage sources shorted. The rating is the converter’s own MVA base; if it is missing the evaluation fails closed rather than inferring a rating. The threshold is $SCR > 3$, taken from the WECC REGC_A strong-grid contract, and the comparison is strict: $SCR \leq 3$ is a refusal.

7.2 The gate is not applicable to this model family

This must be stated before any number, because it is easy to mistake a published diagnostic for an active gate. In ibr_scr_metrics.m:266--276 the screening profile is forced for the project’s full-state dual families:

```
scr_profile = 'not_applicable_full_state_source_model' for model_id ∈
{eecon49_dual, decoupled_dual}
```

The consequence is `eligible_for_scr = false` and `pass = true`, with the recorded reason

“SCR threshold not applicable to this full-state model; equilibrium, limits and full-KCL SSSA remain mandatory”.

[measured] — verified for all four converters on the delivered case: every `per_resource` entry reports `eligible_for_scr = 0`, `SCR = NaN`, `pass = 1`. The reason is sound: the REGC_A screen exists for a reduced current-source plant model, whereas `eecon49_dual` carries the AC filter, the DC link, the PLL/VSG and both control loops as explicit states, so the weak-grid failure the screen protects against is represented in the model rather than assumed away. The SCR number is nevertheless a meaningful *physical* statement about the island, and the run publishes it as a per-sample diagnostic.

7.3 The measured island SCR

Window	n	min	mean	max	below 3
pre-trip, machine online	168	2.0892	2.3439	2.4803	100.00%
post-trip, islanded	2760	0.6501	2.3015	2.8147	100.00%
after the load step	3152	1.5411	2.6899	3.2908	95.53%
fault at bus 9, 85.00–85.15	164	0.2944	1.2954	6.6726	97.56%
line 6–13 out	2160	2.0718	2.7319	6.6184	97.96%
restore to reclose	4384	1.8361	2.5914	3.5322	99.64%
after reclose	1928	1.9104	2.3291	2.4833	100.00%
whole run, all four converters	14716	0.2944	2.5271	6.6726	98.61%

Table 8: Per-sample SCR at the converter buses, from the run’s own `agsi_reference.scr` diagnostic. Classification ASSUMED_DIAGNOSTIC; the ratio uses each converter’s rated MVA on the system base (100 MVA), and the numerator uses the measured bus voltage of that sample. **[measured]**

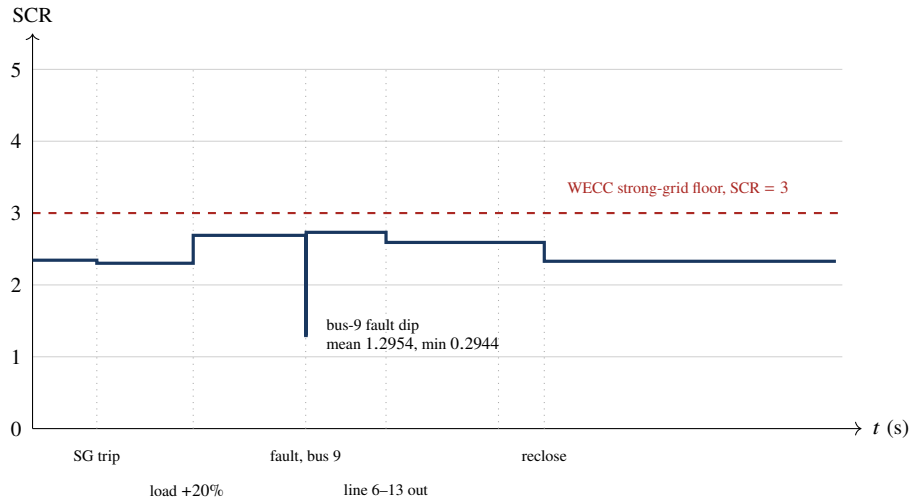


Figure 3: The mean SCR across the four converters, as a step plot of the per-window means of Table 8. The dashed line is the WECC strong-grid floor. The island never leaves the weak-grid region, and the deepest excursion coincides with the fault, not with the machine trip.

7.4 Reading the number

1. **The machine trip barely moves the SCR.** $2.3439 \rightarrow 2.3015$ on the window mean. This is not an error: Z_{th} in (7) is built from the passive network, and the machine is not a term in it. What changes at the trip is the numerator — the measured bus voltage — and, more importantly, *which sources are behind the bus*. Before the trip the converters are followers on a network held up by a 615 MVA machine; after it they are the only sources on the same network. The impedance seen is the same; the support behind it is not.
2. **Everything on this island is weak by the WECC definition.** The 98.61% figure is the striking one: for all but 1.39% of the delivered run’s converter-samples, the short-circuit

ratio at the converter bus is at or below the floor that REGC_A uses to *refuse* a grid-following plant. The deepest value, 0.2944, occurs during the fault, where a collapsed bus voltage is the whole of the numerator. Two further rows of Table 8 sharpen it: the whole post-trip window and the whole post-reclose window are 100% below the floor, and no window’s mean reaches it.

3. **The load step raises SCR.** The window mean rises from 2.3015 to 2.6899 after the +20% step. Load admittance is excluded from $\mathbf{Y}$ but the load current raises bus voltage, and $|V|^2$ is the numerator. This is a reminder that (7) as implemented is a voltage-referenced quantity, not a pure network invariant, and that comparisons should be made within a window rather than across one.
4. **A rating-convention caveat, stated so the number is not over-read.** The separate diagnostic in `build_ieee14_switch_system.m:159` computes the same $|V|^2/|Z_{th}|$ but divides by $\max(|S_{bus}|, 0.20)$ rather than by the converter rating, and reports 4.78–7.19 for the same four buses with the machine online. Same impedance, different denominator. The number quoted in Table 8 is the one the running supervisor’s own diagnostic produces, on the rated-MVA convention.

8 What happens if all four converters form

The comparison study runs six arms from one shared option set: the delivered adaptive policy, three arms that pin the forming set to a fixed cardinality for the whole islanded window, an arm with adaptation disabled, and an arm in which every converter is held grid-following. The study states its own purpose for the pinned arm in the runner header:

“pinned_gfm4: all four pinned. More grid-forming capacity than the small-signal optimum, included because ‘more is safer’ is the intuition this study is testing.” (`run_ieee14_gfm_lock_comparison`)

Arm	horizon reached	samples	termination
locked_gfl	20.0000 s	43	noVoltageFormingSource
pinned_gfm4	25.4852 s	837	adaptiveDtMin, $\Delta t_{\min} = 6.104 \times 10^{-6}$
no_adaptation	250.0000 s	1565	SUCCESS, reclose 151.1321 s
pinned_gfm1	250.0000 s	1766	SUCCESS, reclose 151.1321 s
pinned_gfm2	250.0000 s	1761	SUCCESS, reclose 154.0244 s
adaptive	250.0000 s	3679	SUCCESS, reclose 159.2519 s

Table 9: The six delivered arms. `pinned_gfm4` pins the forming set to all four converters; it dies 5.49 s after the machine trip, before the load step and before any fault. **[measured]**

8.1 The all-four arm does not survive the island

The terminal state of `pinned_gfm4` is not a marginal loss of damping. In its last accepted sample the four converters have separated in frequency: IBR2 runs at 64.9947 Hz while IBR3, IBR6 and IBR8 run at 59.1383, 59.1809 and 59.1946 Hz — a 5.86 Hz split, with one converter having

run away from the other three. The step controller then reported 398 successive Newton non-convergences and terminated at the minimum step 6.104×10^{-6} s. Reclose was never requested. **[measured]**

That is the answer to “what happens if all four become GFM”, but it is not yet an explanation, because Section 5 certified all four as admissible. Both facts are true, and the resolution is the one recorded in docs/project/defects/2026-08-14-all4-gfm-commit-outside-basin.md.

8.2 Why an admissible set can still fail

The acceptance gate of (4) is a statement about an *equilibrium*. It is established by linearising about the solved operating point, and it says nothing about the trajectory that arrives there.

The defect record measured that gap on a *different* all-four transition from the one that ends the pinned arm: its controlled experiment is on the load-step promotion at $t = 53.4025$ s in the pre-certificate baseline, where the baseline loses synchronism after the commit while the delivered run, with the forward certificate in place, commits a conditioned variant at the same instant and completes 250 s. The two failures are the same *class* — an admitting equilibrium whose arriving state is outside its basin — but they are not the same event, and the record does not root-cause the pinned arm’s death. The experiment re-runs the same inputs from two different initial device states:

Arm of the defect experiment	initial device state	outcome	excursion
ARM A	the state the production run actually arrives with	SLIPS	4358°
ARM B	the certified equilibrium state of the same configuration	SYNC	6.6°

Table 10: The same committed configuration, the same inputs, two initial device states. From the production arrival the island loses synchronism; from the certified equilibrium it is stable. **[measured]**

The carrying coordinate is the incumbent former’s own outer voltage-loop integrators. For IBR2, gfm_xi_Vd/xi_Vq stand at 0.144253 in the production arrival against 0.199658 at the certified equilibrium — a 28% displacement in a device state that no bus quantity reflects. The committed configuration is stable; the *arriving state is outside its basin of attraction*.

Remark 3 (Two failure modes, and they are different). The all-four row of Table 5 fails a test of the trajectory (Section 8.1 and this subsection), while the three-unit rows fail the weaker test of Section 4 (the equilibrium does not exist at all). A policy that asked only “is the destination stable?” would admit all four and lose the island — which is exactly what the pre-certificate baseline did at $t = 53.4025$ s. This is why Stage 3 of the pipeline re-simulates before committing, and why the certificate is a forward trial rather than a second eigenvalue check: no eigenvalue test can see the arriving device state.

8.3 Why all four is worse — three claims, each with its scope

The honest statement is narrower than “four is bad”. Three separate statements, each separately evidenced:

1. **It is the least damped admissible configuration.** $\zeta_{\text{worst}} = 0.1223$ against 0.2549–0.2676 for the other six, a factor of about two. Measured, and a property of the equilibrium (Table 5).

2. **Its arrival is not robust.** Measured, and a property of the trajectory (Table 10). This is the failure mode that actually ended the pinned arm.
3. **It is chosen as a residue, not as a judgement.** At $t = 53.3795$ s the all-four set was the *only* admissible destination (Table 6). The policy did not prefer four over two; it had no alternative.

Against those three, one measured statement points the other way and belongs in the same list, because a report that omits it is not a proof:

In the defect record's load-severity experiment, the all-four configuration rides a 2.0 pu load deficit that the two-former configuration cannot — two formers slip at the same deficit. **[measured]**

So the defensible conclusion is not “fewer formers are better”. It is:

The number of forming converters is not a free parameter and stability is not monotone in it. Both endpoints are wrong: with no former the island dies in 20 s, and with four pinned the island dies in 25.5 s. The count has to be chosen per configuration, from the certified margin and the arrival, and the ranking has to be allowed to prefer two to four when two is certified.

8.4 Is a strong grid not a good thing?

It is, and this island does not have one — which is an argument *for* grid-forming, not against it. The difficulty is that “a strong grid” and “more grid-forming converters” are two different quantities, and four facts keep them apart.

1. **In this project's metric, the forming count does not enter the ratio at all.** Y in (7) is built from branch data, bus shunts and the reference-bus grounding, and from nothing else. The published SCR is therefore invariant to how many converters form. Promoting a converter cannot raise the SCR of the network it sits on, whatever else it achieves.
2. **The measured island is weak on every sample that matters.** 98.61% of converter-samples sit at or below the WECC floor of 3, and the post-trip mean is 2.30. The premise behind the question is correct: this is a weak grid. That is exactly why the island needs voltage-forming sources — and why the interesting question is how many, not whether.
3. **Above the count where no follower remains, the criterion loses its subject.** REGC_A screens a grid-following plant because a PLL needs a stiff bus to lock to. With all four converters forming there is no PLL left to protect, and the screening question is vacuous. The problem has changed kind: it is no longer “can the followers lock?” but “can four voltage-source controllers share current on a network whose impedance none of them can change?”. Section 4 shows that is precisely the question this network answers badly — all four three-unit sharings have no equilibrium, and the four-unit sharing has the thinnest margin.
4. **Strength here is delivered by ownership, not by census.** One former whose voltage loop owns its bus gives that bus a defined voltage. What the other three need is not another owner but a defined voltage at their own buses, delivered by a set that the network can actually solve.

Remark 4 (The intuition is right; the conclusion drawn from it is not). “Weak grid $\Rightarrow$ form more” is sound as far as the first step. It fails at the second, because the count is not the strength. The lower bound is measured and hard: the `locked_gfl` arm, with all four converters following, dies at $t = 20.0000$ s — the trip instant itself — with `noVoltageFormingSource` after 43 samples. Something must form. The upper bound is measured too, and it is not four.

9 Where the present two-term index fails

Stage 1 of the pipeline consumes exactly two quantities, (1): a bus-voltage error and a centre-of-inertia frequency error. The judgement that this is too narrow is correct, and the delivered run measures five distinct ways in which it is. All five are stated against `adaptive_250s.mat`; the first three are structural properties of (1) and hold for every run of this code.

9.1 The frequency term carries no device identity

$$\max_t \left[\max_i J_f(t) - \min_i J_f(t) \right] = 0.000 \times 10^0 \quad \text{over all 3679 samples.} \quad (8)$$

J_f is formed from f_{COI} , which is by construction a single network-wide scalar; every converter reads the same value at every instant. Half of S_i in (1) is therefore a common-mode term that cannot distinguish one converter from another. Its consequences are concrete: the index can answer “is the island disturbed?” but structurally cannot answer “*which* converter is in trouble?” — and that is the second question the supervisor has to answer. Stage 2 covers the gap with a static, pre-computed ranking, which is sound but blind to the actual disturbance. **[measured]**

9.2 It saturates before it is informative

S is clipped at 1, and reaches the clip when $J_V + J_f \geq 2$. With both bases fixed at 0.10 pu and 0.50 Hz, that is a bus-voltage error of about 0.2 pu at nominal frequency. The measured maxima over the delivered run are $J_V^{\max} = 7.671$ and $J_f^{\max} = 2.322$: the index is driven to its ceiling by a voltage error nearly four times the saturation threshold. During the bus-9 fault ($t = 85.0000$ – 85.1500 s) all four converters sit at $S = 1.0000$ for 0.147–0.151 s — only about 1.5 times the 0.10 s dwell. The excursion range over the run is $|V| \in [0.0369, 1.5232]$ pu and $f_{\text{COI}} \in [58.8391, 61.1385]$ Hz.

Two consequences follow. First, in the region where the decision is actually made the index has degenerated from a graded measure into a binary disturbance flag.

The second consequence requires care, because in the delivered run the augment test could not have fired during that fault for a different reason. The trigger is evaluated over the grid-following subset only, `any(sev_values(gfl_mask) >= gamma_on) (ts_simulate_ibr_hybrid.m:863)`; all four converters were already forming throughout the fault, so that subset was empty and the test was *vacuously false*. The dwell was not what prevented a transaction. What the 0.147–0.151 s saturation measures is therefore a property of the index, and its decision consequence is a counterfactual: on an island small enough that a follower remained, the same fault would have asserted the augment condition for about 1.5 times the dwell and promoted converters in response to a transient the correct answer is to ride through. **[measured]** for the saturation, *counterfactual* for the transaction it would have produced.

9.3 It cannot see a device state, and the all-four failure lives there

Section 8 established that the all-four loss of synchronism was carried by a device-internal quantity, gfm_xi_Vd/xi_Vq , displaced 28% between the production arrival and the certified equilibrium. Equation (1) contains no device state. No weighting, no base and no threshold applied to $(|V_i|, f_{COI})$ can detect that displacement, because the displacement is not a function of them. **[measured]**

9.4 The PLL lock margin is measured, published, and not used

The run already publishes a PLL lock index, $J_{lock} = |v_q|/0.10$ on grid-following converters (`agsi_reference_terms.m:200--208`). On the delivered run it exceeds 1 in stretches of 1.3–2.7 s with peaks of 4.6–8.0, repeatedly. The most telling instance is $t = 146.3127\text{--}147.6730$ s, with peaks 7.9894, 8.0431 and 7.8085 on three converters — a PLL running roughly 80 times outside its small-angle linearisation limit, roughly two seconds *before* the voltage/frequency index would drive the same augmentation at 148.2811 s. The supervisor acts, and it acts late, for a reason its own diagnostic had already recorded. **[measured]**

9.5 Current headroom is measured, published, and not used

The other device-level quantity the run publishes is the current magnitude. The measured ratio $|I|/I_{max}$ reaches 1.0047–1.0068 on all four converters during the fault, with the current clamp active; the only other excursion above 0.95 in the whole run is IBR6 at $t = 51.54\text{--}51.71$ s, peaking at 0.9636. The current ceiling is the one hard nonlinearity in the device and the direct cause of the voltage sag the index does observe — but (1) sees the consequence and not the cause, and cannot distinguish “a follower is at its ceiling and needs help” from “a former is at its ceiling and must not be released”. **[measured]**

9.6 Summary of the gap

Defect	Measure	Direction
common-mode J_f	spread exactly 0.000×10^0 across devices, all 3679 samples	cannot attribute
saturation	all four at 1.0000 for 0.147–0.151 s during a fault	false positive
no device state	28% displacement in gfm_xi_Vd/xi_Vq is invisible	false negative
PLL lock unused	peaks 7.99/8.04/7.81 at $t = 146.31\text{--}147.67$ s, action at 148.28 s	late
current headroom unused	$ I /I_{max}$ reaching 1.0068 with the clamp active	cannot attribute

Table 11: Five measured failure modes of (1) on the delivered run.

10 Two additional switching indices: RoCoF and RoCoV

10.1 The design constraint that shapes both

The existing reference module already computes six sub-indices — J_V, J_f, J_R (its rate of change of frequency), J_P, J_{SCR}, J_{lock} (`agsi_reference_terms.m:113`) — and its own decision contract

states that *switching consumes J_V and J_f only; every other term published here is reference-only and enters no gate* (:52--53). The two indices proposed below are both *rates*: the rate of change of frequency (RoCoF) and the rate of change of voltage (RoCoV). Neither is added to the sum in (1), for a reason that is structural rather than stylistic: a rate and a deviation are not exchangeable at any set of weights. Trading 10% of a rate margin against 10% of a deviation margin is not a meaningful operation, and a scalar that performs it will certify configurations in which one of the two is unacceptable.

The pairing is deliberate, and it answers the failure measured in Section 9: (1) saturates during the fault, so in the region where the decision is actually made it has degenerated into a binary disturbance flag. A rate is the quantity that survives that degeneration. It is largest exactly when the deviation is still small and moving fastest, and it returns to zero once the disturbance has settled, at which point the deviation is still at its largest. The pair therefore separates three states that the deviation alone confounds: off-nominal and still moving, off-nominal and settled, and nominal.

Accordingly, each proposed index enters its *own* condition, in parallel with (2), and neither is added to the sum in (1). This is the project's own stated rule, not an invention of this report:

“Short-circuit strength, device current limits, and reference ownership are separate non-tradeable gates, never traded off against each other inside one scalar.” (README.md:84--86)

[proposed]

10.2 Index A — RoCoF: the per-device rate of change of frequency

Definition. For each converter, $\text{RoCoF}_i = df_i/dt$ taken from that converter's own speed state. The quantity already exists inside the reference module: `agsi_reference_terms.m:153` computes it per device as `dfdev(k)`, and :165 then *discards* the per-device value by collapsing every device into the inertia-weighted centre-of-inertia scalar that is published as J_R . The proposal is not a new measurement; it is to stop discarding the per-device one.

Base. 1 Hz/s, the project's declared `agsi_rocof_base_Hz_s`. That base is classified `ASSUMED_DIAGNOSTIC` and caller-overridable, and the reference module says so in its own `base_classification` field. The construction used for the other bases in this report would give $df_{\text{base}}/T_{d,\text{on}} = 0.50/0.10 = 5$ Hz/s, so the declared base is the more sensitive of the two. The provenance gap is stated rather than hidden. [proposed]

Scope, and why it is not the augment trigger. Per-device frequency is recorded as NaN on 100.0% of grid-following samples and on 0.0% of grid-forming samples [measured] — the converter speed state exists exactly and only while the machine is forming, because a grid-following branch in this model carries no swing state. The index is therefore defined on the *forming set* alone. It cannot request a promotion, and it is not proposed to.

Measured evidence. Three results, all on the delivered run.

1. **It names the converter that took the transition shock.** At the 1 Hz/s base with the project's on-dwell there are eight qualified excursions in the run, and *seven of the eight begin within 2 ms of a transaction the supervisor made*. At each of the three promotions the unit that fires is one of the units just promoted: IBR6 at $t = 22.0888$ s (promotion at

22.0887 s), IBR6 at $t = 51.3404$ s (promotion at 51.3403 s), IBR3 and IBR8 at $t = 53.3797$ s (promotion at 53.3795 s), IBR6 at $t = 148.2812$ s (promotion at 148.2811 s). The eighth is IBR2, the sole remaining former, at the release of 146.2782 s. **[measured]**

2. **The published scalar cannot substitute for it.** Over the same eight windows the device peak ranges from $0.13\times$ to $2.01\times$ the centre-of-inertia peak — neither bounds the other. The failure is worst exactly at a promotion: at $t = 53.3795$ s the published J_R peaks at 123.04 Hz/s while the two *entering* units reach only 26.16 and 15.95 Hz/s, because the inertia weighting is dominated by the incumbents that did not transition. The scalar reports the incumbents' shock and is silent about the entrants'. **[measured]**
3. **At the fault clearing it separates the formers.** At $t = 85.1500$ s the four forming converters reach -1050.98 , -1016.92 , -1106.62 and -1212.91 Hz/s — a spread of 196.0 Hz/s, 17.9% of the mean magnitude, naming IBR8 as the most shocked. The published J_R has a spread of exactly 0.000×10^0 across converters at every one of the 3679 samples. **[measured]**

Proposed rule. RoCoF does not create a trigger. It enters as a *release guard*, and as the transaction's attribution:

$$\text{block release if } \exists i \in \text{GFM} : \text{RoCoF}_i \geq 1 \text{ Hz/s,} \quad \text{and record } \arg \max_i |\text{RoCoF}_i| \text{ per transaction.} \quad (9)$$

The direction is safe by construction: the index exists only on forming converters, so it can never request a promotion, and blocking a release is the conservative direction.

Validation route. Replay the transaction log and confirm (a) that (9) would have blocked none of the three releases the adaptive policy actually made — measured, all three occurred at $|\text{RoCoF}| \leq 0.066$ Hz/s, so none is blocked — and (b) that the recorded attribution names the promoted unit at each of the three promotions.

10.3 Index B — RoCoV: the rate of change of voltage

Definition. $\text{RoCoV}_i = d|V_i|/dt$ at the converter's own bus.

Base, derived rather than tuned. The reference module has no base for this quantity. This report derives one the same way the project derives its others — as the rate at which the deviation base would be traversed within one on-dwell:

$$\text{RoCoV}_{\text{base}} = \frac{\text{severity_dV_base}}{\text{severity_T_d_on}} = \frac{0.10 \text{ pu}}{0.10 \text{ s}} = 1.0 \text{ pu/s.} \quad (10)$$

Scope. Bus voltage magnitude is recorded with a NaN fraction of 0.00% over all 3679 samples **[measured]** — it exists in both modes and for every converter, unlike RoCoF. RoCoV therefore attributes for followers as well as formers, and it is the only one of the two that *could* serve the augment direction.

Measured evidence. Two results decide its role.

1. **It is a qualifier, not a trigger.** At the bus-9 fault clearing the rate reaches 998.2, 975.7, 1028.5 and 1026.7 pu/s on the four converters (p_{99} over the run is 81.7–86.6 pu/s), yet the *longest continuous excursion above the derived base anywhere in the run is 0.0369 s* — 0.37 times the 0.10 s on-dwell. With the project’s dwell, RoCoV has *zero* qualified excursions on this run. A rule that paired RoCoV with $T_{d,on}$ would be inert, and the measurement says so before anyone implements it. **[measured]**
2. **It identifies a 35-second decision that the two-term index cannot resolve.** Over $t = 110.0021 \rightarrow 145.0000$ s — 34.998 s — the release test was blocked because some S_i stood at up to 0.4301, above the release threshold of 0.35. Over that same stretch every bus had stopped moving: the largest $|\text{RoCoV}|$ anywhere on the island was 0.6451 pu/s, below the derived base. The deviation was a *settled* offset from the frozen pre-event reference, produced by the line 6–13 trip at $t = 110$ s, not an evolving stress. Across the whole run 2239 of 3679 samples (60.9%) have some $S_i \geq 0.35$ while every bus is settled. The cause is structural: J_V is referenced to a frozen pre-event healthy voltage (agsi_reference_terms.m:183--188), so any topology change leaves a permanent offset that the index cannot distinguish from stress. RoCoV is the only published quantity that separates the two, and the two-term index has no way to recover the distinction on its own. **[measured]**

Proposed rule. RoCoV does not create a trigger either. It qualifies the release test and the reference of the deviation term:

$$\text{a sample counts toward the release dwell only if } \max_i |\text{RoCoV}_i| < 1.0 \text{ pu/s}, \quad (11)$$

and a stretch that is settled while off-nominal is *re-referenced* — the frozen healthy voltage is replaced by the settled value — instead of being carried indefinitely as stress. Both halves are structural changes to what the deviation term is measured against; neither changes the 0.35 threshold, and this report does not propose re-tuning that threshold.

Validation route. Replay the run and confirm (a) that no release is granted while any bus is moving faster than the base — measured, the three releases occurred at $|\text{RoCoV}| \leq 0.019$ pu/s, so none is affected — and (b) that the long settled stretches listed above collapse to re-reference events rather than to a blocked release.

10.4 What is deliberately not proposed

J_{SCR} (measured max 9.191) and J_P (measured max 6.158) remain reference-only. J_{SCR} is a network invariant that, as Section 8.4 established, does not respond to the forming count at all, so it cannot discriminate between the destinations a promotion decision has to choose among. J_P measures tracking error against a reference the supervisor itself commands, so it reports the consequence of a mode decision rather than an independent state of the plant. Promoting either would add a term without adding a distinction. **[proposed]**

Remark 5 (Total effect on the decision). Neither proposal changes *when* an augmentation is requested, and neither touches the acceptance criterion of (4). RoCoF (A) guards the release of a former and attributes the transition shock to a named converter. RoCoV (B) qualifies the release dwell and re-references a settled voltage offset. Both act on the handback side of the

policy — the side that Section 6 showed is deliberately the more conservative of the two, and the side on which the delivered run held the least-damped admissible configuration for 92.90 s, of which a measured 34.998 s was a settled offset rather than a live stress.

11 What this report does not claim

1. **It does not claim the delivered policy is optimal.** It claims it implements Proposition 1, and proves that Proposition 1 has the consequences observed. A policy that ranked differently and stayed inside the same admissibility gates is not excluded.
2. **It does not claim two formers are always better than four.** The defect record contains a measured case where all four rides a 2.0 pu deficit that two formers cannot. The claim is that stability and robustness are not monotone in the count, both directions.
3. **It does not claim the SCR values are plant-strength ratings.** Equation (7) is voltage-referenced as implemented, the gate is exempt for this model family, and a second diagnostic in the repository gives 4.78–7.19 for the same buses under a different rating convention. Comparisons are valid within Table 8’s convention.
4. **It does not claim the admissibility table is load-independent.** Table 5 is the frozen base-load table. The +20% load findings quoted in Section 8 are a separate measurement from the defect record, at a different load level.
5. **It does not claim the two indices are implemented.** They are a design proposal, each with a stated validation route and no measured performance. No result in this report depends on them.
6. **It does not claim the basin statement is general.** The 4358° against 6.6° contrast is measured on one transition pair; the mechanism is offered as an explanation of that pair, not as a law.

12 Reproduction

The delivered result set and the report build. All commands are run from the repository root.

```
% 1. the six-arm comparison that produces the shared admissibility table
matlab -batch "run('scripts/reporting/run_ieee14_gfm_lock_comparison.m')"

% 2. the English report
cd docs/source
xelatex report_ieee14_switching_decision_basis_en.tex
xelatex report_ieee14_switching_decision_basis_en.tex
```

The regression gate for the delivered arm is the seven-value row of Table 2: horizon 250.000000 s, 3679 samples, 161 rejected steps, maximum accepted KCL residual 9.972081×10^{-9} , terminal f_{COI} 60.000001128 Hz, reclose time 159.2519 s with status SUCCESS, and hand-back completion at 173.1271 s. If a re-run does not reproduce all seven, the numbers in this report do not apply to it.